① $\boxed{A + A\bar{A}}$

$A + \underbrace{A\bar{A}}_{0} = A + 0 = A$ //

② $\boxed{A + B + \bar{B}}$

$A + \underbrace{B + \bar{B}}_{1} = A + 1 = 1$ //

③ $\boxed{A + B\bar{B}}$

$A + \underbrace{B\bar{B}}_{0} = A + 0 = A$ //

④ $\boxed{AB\bar{B}}$

$A\underbrace{B\bar{B}}_{0} = A.0 = 0$ //

⑤ $\boxed{A(B + \bar{B})}$

$A\underbrace{(B + \bar{B})}_{1} = A.1 = A$ //

⑥ $\boxed{x + \bar{x}y}$

$x + \bar{x}y$

$= x.1 + \bar{x}y$

$= x(1+y) + \bar{x}y$

$= x + xy + \bar{x}y$

$= x + y\underbrace{(x + \bar{x})}_{1}$

$= x + y$

$(x + \bar{x}y) = x + y$ (boolean kuralı)

⑦ $\boxed{AB + \overline{AB}C}$

$\underbrace{AB}_{x} + \underbrace{\overline{AB}}_{\bar{x}}\underbrace{C}_{y}$

$x + \bar{x}y = x + y$ → boolean kuralı

$AB + C$ //

⑧ $\boxed{A + \bar{\bar{A}}}$

$A + \underbrace{\bar{\bar{A}}}_{A} = A + A = A$ //

⑨ $\boxed{A + \overline{\overline{AB}}}$

$A + \underbrace{\overline{\overline{AB}}}_{AB} = A + AB$

$= A\underbrace{(1+B)}_{1}$

$= A$ //

⑩ $\boxed{\overline{\overline{A(B + \bar{B})}}}$

$\overline{\overline{A\underbrace{(B + \bar{B})}_{1}}}$

$= \overline{\overline{A.1}}$

$= \bar{\bar{A}}$

$= A$ //

⑪ $\boxed{A\bar{A} + B\bar{B}}$

$\underbrace{A\bar{A}}_{0} + \underbrace{B\bar{B}}_{0} = 0 + 0 = 0$ //

⑫ $\boxed{(A + \bar{A}).B.\bar{B}}$

$\underbrace{(A + \bar{A})}_{1} . \underbrace{B\bar{B}}_{0}$

$1.0 = 0$ //

⑬ $\boxed{A\bar{A} + AA + BB}$

$\underbrace{A\bar{A}}_{0} + \underbrace{AA}_{A} + \underbrace{BB}_{B}$

$0 + A + B = A + B$ //

⑭ $\boxed{A\bar{A} + AA + B\bar{B}}$

$\underbrace{A\bar{A}}_{0} + \underbrace{AA}_{A} + \underbrace{B\bar{B}}_{0}$

$0 + A + 0 = A$ //

15) $\boxed{C\bar{C}+(A+\bar{A})+(B+\bar{B})+D}$

$$\underbrace{C\bar{C}}_{0}+\underbrace{(A+\bar{A})}_{1}+\underbrace{(B+\bar{B})}_{1}+D$$

$$0+1+1+D=1 //$$

16) $\boxed{ABC+\overline{ABC}\,D}$

$$\underbrace{ABC}_{x}+\underbrace{\overline{ABC}}_{\bar{x}}\,\underbrace{D}_{y}$$

$x+\bar{x}y=x+y$ boolean kuralı

$$=ABC+D //$$

17) $\boxed{AB\bar{C}\bar{B}D}$

$$AB\bar{C}\bar{B}D=A\bar{C}\underbrace{B\bar{B}}_{0}.D$$

$$=A\bar{C}.0.D=0 //$$

18) $\boxed{AB+\overline{AB}CD+AB\bar{C}DC}$

$$\underbrace{AB}_{x}+\underbrace{\overline{AB}}_{\bar{x}}\underbrace{CD}_{y}+AB\bar{C}DC$$

$x+\bar{x}y=x+y$ boolean kuralı

$$AB+CD+\underbrace{ABD\underbrace{\bar{C}C}_{0}}_{0}$$

$$=AB+CD //$$

19) $\boxed{C+BC}$

$$C+BC$$

$$C(\underbrace{1+B}_{1})=C.1=C //$$

20) $\boxed{A(A+B)}$

$$A(A+B)=\underbrace{AA}_{A}+AB$$

$$=A+AB$$

$$=A(\underbrace{1+B}_{1})$$

$$=A.1=A //$$

21) $\boxed{\overline{A+B}}$

$$\overline{A+B}$$

$$=\bar{A}.\bar{B} //$$

(de-morgan kuralı)

22) $\boxed{\overline{AB}}$

$$\overline{AB}=\bar{A}+\bar{B}$$

(demorgan kuralı)

23) $\boxed{\overline{A+BC}}$

$$\overline{A+BC}$$

$$\bar{A}.\overline{BC}$$

$$\bar{A}.(\bar{B}+\bar{C})$$

$$\bar{A}\bar{B}+\bar{A}\bar{C} //$$

24) $\boxed{A(\bar{A}+\bar{B})}$

$$A(\bar{A}+\bar{B})$$

$$\underbrace{A\bar{A}}_{0}+A\bar{B}$$

$$0+A\bar{B}$$

$$=A\bar{B} //$$

25) $\boxed{D+\overline{CD}}$

$$D+\overline{CD}$$

$$=D+\bar{C}+\bar{D}$$

$$=\underbrace{D+\bar{D}}_{1}+\bar{C}$$

$$=1+\bar{C}$$

$$=1 //$$

(26) $\boxed{A(A+C)}$

$A(A+C)$

$=\underbrace{AA}_{A}+AC$

$=A+AC$

$=A(\underbrace{1+C}_{1})$

$=A.1=A$ //

(27) $\boxed{A+AB}$

$A+AB$

$=A(\underbrace{1+B}_{1})$

$=A.1$

$=A$ //

(28) $\boxed{A+\bar{A}B}$

$A+\bar{A}B$

$\underbrace{x+\bar{x}y=x+y}$ De-morgan kuralı

$=A+B$

veya

$=A.1+\bar{A}B$

$=A(1+B)+\bar{A}B$

$=A+AB+\bar{A}B$

$=A+B(\underbrace{A+\bar{A}}_{1})$ $=A+B$ //

(29) $\boxed{A(\bar{A}+B)}$

$A(\bar{A}+B)$

$=\underbrace{A.\bar{A}}_{0}+AB$

$=0+AB$

$=AB$ //

(30) $\boxed{A+\bar{A}\bar{B}}$

$\underbrace{A}_{x}+\underbrace{\bar{A}}_{\bar{x}}\underbrace{\bar{B}}_{y}$

$x+\bar{x}y=x+y$: boolean kuralı

$=A+\bar{B}$ //

(31) $\boxed{A+\bar{A}\bar{B}\bar{C}}$

$\underbrace{A}_{x}+\underbrace{\bar{A}}_{\bar{x}}\underbrace{\bar{B}\bar{C}}_{y}$

$x+\bar{x}y=x+y$ boolean kuralı

$=A+\bar{B}\bar{C}$ //

(32) $\boxed{AB+\overline{(A+B)}}$

$AB+\overline{(A+B)}$

$=AB+\bar{A}\bar{B}$ //

(33) $\boxed{C+\overline{BC}}$

$C+\overline{BC}$

$=C+\bar{B}+\bar{C}$

$=\underbrace{C+\bar{C}}_{1}+\bar{B}$

$=1+\bar{B}=1$ //

(34) $\boxed{A\bar{B}+AB+C}$

$A\bar{B}+AB+C$

$=A(\underbrace{\bar{B}+B}_{1})+C$

$=A+C$ //

(35) $\boxed{(X+Z)(XY)}$

$(X+Z)XY$

$=\underbrace{XX}_{X}Y+ZXY$

$=XY+ZXY$

$=XY(\underbrace{1+Z}_{1})$

$=XY$ //

(36) $\boxed{A+\bar{A}B+AB}$

$A+\bar{A}B+AB$

$x+\bar{x}y=x+y$ boolean kuralı

$=A+B+AB$

$=A(\underbrace{1+B}_{1})+B$

$=A+B$ //

(37) $\boxed{A+AB+A\bar{B}C}$

$A+AB+A\bar{B}C$

$=A(\underbrace{1+B+\bar{B}C}_{1})$

$=A.1$

$=A$ //

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(38)

$\boxed{A\bar{C}+AB\bar{C}}$

$A\bar{C}+AB\bar{C}$

$=A\bar{C}+A\bar{C}B$

$=A\bar{C}(\underbrace{1+B}_{1})$

$=A\bar{C}$//

(39)

$\boxed{(A+B)(A+C)}$

$(A+B)(A+C)$

$=\underbrace{AA}_{A}+AC+BA+BC$

$=A+AC+AB+BC$

$=A(\underbrace{1+C+B}_{1})+BC$

$=A.1+BC = A+BC$//

(40)

$\boxed{\bar{A}BC+\bar{A}}$

$\bar{A}BC+\bar{A}$

$=\bar{A}(\underbrace{BC+1}_{1})$

$=\bar{A}.1$

$=\bar{A}$//

(41)

$\boxed{A\bar{B}D+A\bar{B}\bar{D}}$

$A\bar{B}D+A\bar{B}\bar{D}$

$=A\bar{B}(\underbrace{D+\bar{D}}_{1})$

$=A\bar{B}$//

(42)

$\boxed{(\bar{A}+B)(A+B)}$

$(\bar{A}+B)(A+B)$

$=\underbrace{\bar{A}A}_{0}+\bar{A}B+BA+\underbrace{BB}_{B}$

$=0+\bar{A}B+BA+B$

$=\bar{A}B+BA+B$

$=B(\underbrace{\bar{A}+A+1}_{1})$

$=B.1=B$//

(43)

$\boxed{A(B+CD)}$

$A(B+CD)$

$=AB+ACD$//

(44)

$\boxed{AB+\bar{A}C+\bar{B}A}$

$AB+\bar{A}C+\bar{B}A$

$=AB+\bar{B}A+\bar{A}C$

$=A(\underbrace{B+\bar{B}}_{1})+\bar{A}C$

$=A+\bar{A}C$ $(x+\bar{x}y=x+y)$ boolean kuralı

$=A+C$//

(45)

$\boxed{A(\bar{A}+B)+A\bar{B}}$

$A(\bar{A}+B)+A\bar{B}$

$=\underbrace{A\bar{A}}_{0}+AB+A\bar{B}$

$=AB+A\bar{B}$

$=A(\underbrace{B+\bar{B}}_{1})$

$=A$//

(46)

$\boxed{A+B(A+C)+AC}$

$A+B(A+C)+AC$

$=A+BA+BC+AC$

$=A(\underbrace{1+B+C}_{1})+BC$

$=A+BC$//

(47)

$\boxed{AB+ABC}$

$AB+ABC$

$=AB(\underbrace{1+C}_{1})$

$=AB$//

(48)

$\boxed{A+\bar{A}\bar{B}C+BC\bar{D}}$

$\underbrace{A}_{x}+\underbrace{\bar{A}}_{\bar{x}}\underbrace{\bar{B}C}_{y}+BC\bar{D}$

$x+\bar{x}y=x+y$ boolean kura

$=A+\bar{B}C+BC\bar{D}$

$=A+C(\underbrace{\bar{B}}_{x}+\underbrace{B}_{\bar{x}}\underbrace{\bar{D}}_{y})$

$=A+C(\bar{B}+\bar{D})$

$=A+\bar{B}C+C\bar{D}$//

(49) $\boxed{AB+BC(B+C)}$

$AB+BC(B+C)$

$=AB+BC\cancel{B}+BC\cancel{C}$

$=AB+\underbrace{BC+BC}_{BC}$

$=AB+BC$ //

veya $=A(B+C)$//

(50) $\boxed{\overline{(A+B).C}}$

$\overline{(A+B).C}$

$=\overline{A+B}+\overline{C}$

$=\overline{A}\,\overline{B}+\overline{C}$//

(51) $\boxed{\overline{A+B+C}}$

$\overline{A+B+C}$

$=\overline{A}\,\overline{B}\,\overline{C}$ // (de-morgan)

(52) $\boxed{\overline{A}\,\overline{B}\,\overline{C}+\overline{A}\,\overline{B}C+AB\overline{C}+A\overline{B}\,\overline{C}}$

$\overline{A}\,\overline{B}\,\overline{C}+\overline{A}\,\overline{B}C+AB\overline{C}+A\overline{B}\,\overline{C}$

$=\overline{A}\,\overline{B}(\underbrace{\overline{C}+C}_{1})+A\overline{C}(\underbrace{B+\overline{B}}_{1})$

$=\overline{A}\,\overline{B}+A\overline{C}$//

(53) $\boxed{A+\overline{A}B+\overline{A}C+CD}$

$\underset{x}{A}+\underset{\overline{x}}{\overline{A}}\underset{y}{B}+\overline{A}C+CD$

$x+\overline{x}y=x+y$ boolean kuralı

$=A+B+\overline{A}C+CD$

$=\underset{x}{A}+\underset{\overline{x}}{\overline{A}}\underset{y}{C}+B+CD$

$=A+C+B+CD$//

(54) $\boxed{\overline{A}\,\overline{B}C+\overline{A}BC+A\overline{B}}$

$\overline{A}\,\overline{B}C+\overline{A}BC+A\overline{B}$

$=\overline{A}C(\underbrace{\overline{B}+B}_{1})+A\overline{B}$

$=\overline{A}C+A\overline{B}$//

(55) $\boxed{(\overline{A}+B)(A+C)(B+C)}$

$(\overline{A}+B)(A+C)(B+C)$

$=(\underbrace{\overline{A}A}_{0}+AB+BC)(B+C)$

$=(AB+BC)(B+C)$

$=AB\cancel{B}+ABC+BC\cancel{B}+BC\cancel{C}$

$=AB+ABC+BC+\cancel{BC}$

$=AB(\underbrace{1+C}_{1})+BC$

$=AB+BC$//

(56) $\boxed{\overline{A}B+BC+AC}$

$\overline{A}B+BC+AC$

$=\overline{A}BC+\overline{A}B\overline{C}+ABC+\cancel{\overline{A}BC}+\cancel{ABC}+A\overline{B}C$

(standart çarpımların toplamı)

$=\overline{A}BC+\overline{A}B\overline{C}+ABC+A\overline{B}C$

$=\overline{A}B(\underbrace{C+\overline{C}}_{1})+AC(\underbrace{B+\overline{B}}_{1})$

$=\overline{A}B+AC$//

57

$\boxed{\overline{A}B + A + AB}$

$\overline{A}B + A + AB$

$= \overline{A}B + A(\underbrace{1+B}_{1})$

$= \overline{A}B + A \quad = A + \overline{A}B$

$x + \overline{x}y = x + y$ boolean kuralı

$= A + B$ //

58

$\boxed{AB + \overline{A}B + \overline{A}\overline{B}}$

$AB + \overline{A}B + \overline{A}\overline{B}$

$= B(\underbrace{A+\overline{A}}_{1}) + \overline{A}\overline{B}$

$= \underset{x}{B} + \underset{\overline{x}}{\overline{B}}\,\underset{y}{\overline{A}}$

$x + \overline{x}y = x + y$ boolean kuralı

$= B + \overline{A}$ //

59

$\boxed{AB + A\overline{B} + \overline{A}C + \overline{A}\overline{C}}$

$AB + A\overline{B} + \overline{A}C + \overline{A}\overline{C}$

$= A(\underbrace{B+\overline{B}}_{1}) + \overline{A}(\underbrace{C+\overline{C}}_{1})$

$= A + \overline{A}$

$= 1$ //

60

$\boxed{AB + AB\overline{C} + AB\overline{C} + ABC}$

$AB + AB\overline{C} + \cancel{AB\overline{C}} + ABC$

$= AB + AB\overline{C} + ABC$

$= AB(\underbrace{1+\overline{C}+C}_{1})$

$= AB$ //

61

$\boxed{AB(A+B)(B+B)}$

$AB(A+B)(\underbrace{B+B}_{B})$

$= AB(A+B) \cdot B$

$= A\underbrace{BB}_{B}(A+B)$

$= AB(A+B)$

$= AB\cancel{A} + AB\cancel{B}$

$= AB + AB = AB$ //

62

$\boxed{AB + BC + AC}$

$AB + BC + AC$

Daha fazla sadeleşmez.

63

$\boxed{ABC + AB\overline{C} + \overline{A}B\overline{C}}$

$ABC + AB\overline{C} + \overline{A}B\overline{C}$

$= AB(\underbrace{C+\overline{C}}_{1}) + \overline{A}B\overline{C}$

$= AB + \overline{A}B\overline{C}$

$= B(A + \overline{A}\overline{C}) \quad x + \overline{x}y = x + y$ boolean kuralı

$= B(A + \overline{C})$

$= AB + B\overline{C}$ //

64

$\boxed{A\overline{C} + AB\overline{C}}$

$A\overline{C} + AB\overline{C}$

$= A\overline{C}(\underbrace{1+B}_{1})$

$= A\overline{C}$ //

65.

$$\boxed{A\bar{B}D + A\bar{B}\bar{D}}$$

$$A\bar{B}D + A\bar{B}\bar{D}$$

$$= A\bar{B}(\underbrace{D+\bar{D}}_{1})$$

$$= A\bar{B} \;//$$

66.

$$\boxed{C\,\overline{(A+B)} + D}$$

$$C\,\overline{(A+B)} + D$$

$$= C(\bar{A}\bar{B}) + D$$

$$= \bar{A}\bar{B}C + D \;//$$

67.

$$\boxed{(\bar{A}+B)[\bar{A}(B+A)]}$$

$$(\bar{A}+B)[\bar{A}(B+A)]$$

$$= (\bar{A}+B)[\bar{A}B + \underbrace{\bar{A}A}_{0}]$$

$$= (\bar{A}+B)\cdot(\bar{A}B)$$

$$= \bar{A}\cancel{\bar{A}}B + B\bar{A}\cancel{B}$$

$$= \bar{A}B + B\bar{A} = \bar{A}B \;//$$

68.

$$\boxed{ABC + A(\bar{B}+\bar{C})}$$

$$ABC + A(\bar{B}+\bar{C})$$

$$= ABC + A\bar{B} + A\bar{C}$$

$$= A(BC+\bar{B}) + A\bar{C}$$

69.

$$\boxed{(A+C)(AD+A\bar{D}) + AC + C}$$

$$(A+C)(AD+A\bar{D}) + AC + C$$

$$= (A+C)[A(\underbrace{D+\bar{D}}_{1})] + C(\underbrace{A+1}_{1})$$

$$= (A+C)\cdot A + C$$

$$= AA + AC + C$$

$$= A + AC + C = A(\underbrace{1+C}_{1}) + C$$

$$= A + C \;//$$

70.

$$\boxed{A(A+B) + (B+AA)(A+B)}$$

$$A(A+B) + (B+\underbrace{AA}_{A})(A+B)$$

$$= A(A+B) + (B+A)(A+B)$$

$$= A(A+B) + \underbrace{(A+B)(A+B)}_{A+B}$$

$$= A(A+B) + (A+B)$$

$$= (A+B)(\underbrace{A+1}_{1})$$

$$= A+B \;//$$

71.

$$\boxed{\bar{A}B(C+\bar{C}) + BC(A+\bar{A}) + AC(B+\bar{B})}$$

$$\bar{A}B(\underbrace{C+\bar{C}}_{1}) + BC(\underbrace{A+\bar{A}}_{1}) + AC(\underbrace{B+\bar{B}}_{1})$$

$$= \bar{A}B + BC + AC$$

$$= \underbrace{\bar{A}BC + \bar{A}B\bar{C}}_{\bar{A}B} + \underbrace{ABC + \cancel{\bar{A}BC}}_{BC} + \underbrace{A\cancel{B}C + A\bar{B}C}_{AC}$$

standart çarpımların toplamına dönüştürdük

$$= \bar{A}BC + \bar{A}B\bar{C} + ABC + A\bar{B}C$$

$$= \bar{A}B(\underbrace{C+\bar{C}}_{1}) + AC(\underbrace{B+\bar{B}}_{1})$$

$$= \bar{A}B + AC \;//$$

72.

$$\boxed{\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC}$$

$$\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$= BC(\underbrace{\bar{A}+1}_{1}) + A\bar{B}C + AB\bar{C}$$

$$= BC + A\bar{B}C + AB\bar{C}$$

$$= C(\underbrace{B+A\bar{B}}_{B+A}) + AB\bar{C}$$

$$= CB + CA + AB\bar{C} = B(\overbrace{C+A\bar{C}}^{C+A}) + AC$$

$$= BC + BA + AC \;//$$

(73)

$$\boxed{ABC\left[ABC+\bar{C}(BC+AC)\right]}$$

$$ABC\left[ABC+\bar{C}(BC+AC)\right]$$

$$=ABC\left[ABC+\underbrace{BC\bar{C}}_{0}+\underbrace{AC\bar{C}}_{0}\right]$$

$$=ABC\cdot ABC=ABC//$$

(74)

$$\boxed{ABC+A\bar{B}(\bar{A}\bar{C})}$$

$$ABC+A\bar{B}(\bar{A}\bar{C})$$

$$=ABC+A\bar{B}\bar{A}\bar{C}$$

$$=ABC+\underbrace{A\bar{A}}_{0}\bar{B}\bar{C}=ABC//$$

(75)

$$\boxed{ABC+A\bar{B}(\overline{\bar{A}\bar{C}})}$$

$$ABC+A\bar{B}(\overline{\bar{A}\bar{C}})$$

$$=ABC+A\bar{B}(\bar{\bar{A}}+\bar{\bar{C}})$$

$$=ABC+A\bar{B}(A+C)$$

$$=ABC+A\bar{B}A+A\bar{B}C$$

$$=ABC+A\bar{B}+A\bar{B}C$$

$$=ABC+A\bar{B}(\underbrace{1+C}_{1})$$

$$=ABC+A\bar{B}$$

$$=A(\underbrace{BC+\bar{B}}_{\bar{B}+C})$$

$x+\bar{x}y=x+y$ boolean kuralı

$$=A\bar{B}+AC//$$

(76)

$$\boxed{\overline{AB}\;\overline{(A+B)}\cdot C}$$

$$\overline{AB}\;\overline{(A+B)}\cdot C$$

$$=(\bar{A}+\bar{B})\cdot(\bar{A}\bar{B})C$$

$$=(\bar{A}+\bar{B})\cdot\bar{A}\bar{B}C$$

$$=\bar{A}\bar{A}\bar{B}C+\bar{A}\bar{B}\bar{B}C$$

$$=\bar{A}\bar{B}C+\bar{A}\bar{B}C$$

$$=\bar{A}\bar{B}C//$$

(77)

$$\boxed{\bar{A}\bar{B}+\bar{C}A+\bar{\bar{C}}B}$$

$$\bar{A}\bar{B}+\bar{C}A+\bar{\bar{C}}B$$

$$=\bar{A}\bar{B}+A\bar{C}+BC//$$

(78)

$$\boxed{\overline{A+\bar{B}}+\bar{C}}$$

$$\overline{A+\bar{B}}+\bar{C}$$

$$=\bar{A}\cdot\bar{\bar{B}}+\bar{C}$$

$$=\bar{A}B+\bar{C}//$$

(79)

$$\boxed{(A+\bar{B}+\bar{C})(A+\bar{B}C)}$$

$$(A+\bar{B}+\bar{C})(A+\bar{B}C)$$

$$=AA+A\bar{B}C+A\bar{B}+\bar{B}\bar{B}C+A\bar{C}+\bar{B}\underbrace{C\bar{C}}_{0}$$

$$=A+A\bar{B}C+A\bar{B}+\bar{B}C+A\bar{C}$$

$$=A(\underbrace{1+\bar{B}C+\bar{B}+\bar{C}}_{1})+\bar{B}C=A+\bar{B}C//$$

(80)

$$\boxed{(\bar{A}+B+C)(\bar{B}+C+\bar{D})(\bar{A}+\bar{B}+\bar{C}+D)}$$

$$(\bar{A}+B+C)(\bar{B}+C+\bar{D})(\bar{A}+\bar{B}+\bar{C}+D)$$

$$=(\bar{A}+B+C)(\bar{B}\bar{A}+\underbrace{\bar{B}\bar{B}}_{\bar{B}}+\bar{B}\bar{C}+\bar{B}D+\bar{A}C+\bar{B}C+\underbrace{C\bar{C}}_{0}+CD+\bar{A}\bar{D}+\bar{B}\bar{D}+\bar{C}\bar{D}+\underbrace{\bar{D}D}_{0})$$

$$=(\bar{A}+B+C)(\bar{B}\bar{A}+\bar{B}+\bar{B}\bar{C}+\bar{B}D+\bar{A}C+\bar{B}C+CD+\bar{A}\bar{D}+\bar{B}\bar{D}+\bar{C}\bar{D})$$

$$=(\bar{A}+B+C)[\bar{B}(\underbrace{\bar{A}+1+\bar{C}+D+\bar{D}}_{1})+\bar{A}C+CD+\bar{A}\bar{D}+\bar{C}\bar{D})$$

$$=(\bar{A}+B+C)(\bar{B}+\bar{A}C+CD+\bar{A}\bar{D}+\bar{C}\bar{D})$$

$$=\bar{A}\bar{B}+\bar{A}\cancel{\bar{A}}C+\bar{A}CD+\bar{A}\cancel{\bar{A}}\bar{D}+\bar{A}\bar{C}\bar{D}+\underbrace{B\bar{B}}_{0}+B\bar{A}C+BCD+B\bar{A}\bar{D}+B\bar{C}\bar{D}+C\bar{B}+C\bar{A}\cancel{C}+C\cancel{C}D+C\bar{A}\bar{D}+\underbrace{C\bar{C}\bar{D}}_{0}$$

$$=\bar{A}\bar{B}+\underbrace{\bar{A}C+\bar{A}CD}_{\bar{A}C}+\underbrace{\bar{A}\bar{D}+\bar{A}\bar{C}\bar{D}}_{\bar{A}\bar{D}}+\bar{A}BC+BCD+\bar{A}B\bar{D}+B\bar{C}\bar{D}+\bar{B}C+\cancel{\bar{A}C}+\underbrace{CD+\bar{A}C\bar{D}}_{C(\underbrace{D+\bar{D}\bar{A}}))}$$

$$C(D+\bar{A})$$

$$CD+\bar{A}C$$

$$=\bar{A}\bar{B}+\bar{A}C+\bar{A}\bar{D}+\bar{A}BC+BCD+\bar{A}B\bar{D}+B\bar{C}\bar{D}+\bar{B}C+CD+\cancel{\bar{A}C}$$

$$BCD,\ CD \rightarrow CD$$

$$=\bar{A}B+\bar{A}C+\bar{A}\bar{D}+\bar{A}BC+CD+\bar{A}B\bar{D}+B\bar{C}\bar{D}+\bar{B}C$$

$$\bar{A}B,\ \bar{A}BC \rightarrow \bar{A}B \qquad \bar{A}\bar{D},\ \bar{A}B\bar{D} \rightarrow \bar{A}\bar{D}$$

$$=\bar{A}B+\bar{A}C+\bar{A}\bar{D}+CD+B\bar{C}\bar{D}+\bar{B}C\ //$$

(81)

$$\boxed{(\bar{A}+B)(\bar{B}C+B\bar{C})(A+C)(\bar{C}+A)}$$

$$(\bar{A}+B)(\bar{B}C+B\bar{C})(A+C)(\bar{C}+A)$$

$$=(\bar{A}+B)(\bar{B}C+B\bar{C})(A\bar{C}+A\cancel{A}+\underbrace{C\bar{C}}_{0}+CA)$$

$$=(\bar{A}+B)(\bar{B}C+B\bar{C})(A\bar{C}+A+CA)$$

$$=(\bar{A}+B)(\bar{B}C+B\bar{C})\underbrace{[A(\underbrace{\bar{C}+1+C}_{1})]}_{A}$$

$$=A(\bar{A}+B)(\bar{B}C+B\bar{C})$$

$$=(\underbrace{A\bar{A}}_{0}+AB)(\bar{B}C+B\bar{C})=AB(\bar{B}C+B\bar{C})=\underbrace{AB\underbrace{\bar{B}}_{0}C}_{0}+AB\cancel{B}\bar{C}=AB\bar{C}\ //$$

82

$$\boxed{\overline{B}+\overline{B}C+AB\overline{C}}$$

$$\overline{B}+\overline{B}C+AB\overline{C}$$

$$=\overline{B}(\underbrace{1+C}_{1})+AB\overline{C}$$

$$=\overline{B}+AB\overline{C} \quad =\underbrace{\overline{B}}_{x}+\underbrace{B}_{\overline{x}}\underbrace{A\overline{C}}_{y}$$

$x+\overline{x}y=x+y$ boolean kuralı

$$=\overline{B}+A\overline{C} \;//$$

83

$$\boxed{AB+B\overline{C}+\overline{B}C+\overline{A}B}$$

$$AB+B\overline{C}+\overline{B}C+\overline{A}B$$

$$=B(\underbrace{A+\overline{A}}_{1})+B\overline{C}+\overline{B}C$$

$$=B+B\overline{C}+\overline{B}C \quad =B(\underbrace{1+\overline{C}}_{1})+\overline{B}C$$

$$=\underbrace{B}_{x}+\underbrace{\overline{B}}_{\overline{x}}\underbrace{C}_{y}$$

$x+\overline{x}y=x+y$ kuralı

$$=B+C \;//$$

84

$$\boxed{\overline{A(A+C)}}$$

$$\overline{A(A+C)}$$

$$=\overline{AA+AC} \quad =\overline{A+AC}$$

$$=\overline{A(\underbrace{1+C}_{1})}$$

$$=\overline{A} \;//$$

85

$$\boxed{\overline{A}(A+B)+(B+AA)(A+\overline{B})}$$

$$\overline{A}(A+B)+(B+AA)(A+\overline{B})$$

$$=\underbrace{\overline{A}A}_{0}+\overline{A}B+(B+A)(A+\overline{B})$$

$$=\overline{A}B+BA+\underbrace{B\overline{B}}_{0}+\underbrace{AA}_{A}+A\overline{B}$$

$$=\overline{A}B+AB+A+A\overline{B} \quad =\overline{A}B+A(\underbrace{B+1+\overline{B}}_{1})$$

$$=\overline{A}B+A \quad (x+\overline{x}y=x+y \text{ kuralı})$$

$$=A+B \;//$$

86

$$\boxed{\overline{AB}(\overline{A}+B)(\overline{B}+B)}$$

$$\overline{AB}(\overline{A}+B)\underbrace{(\overline{B}+B)}_{1}$$

$$=(\overline{A}+\overline{B})(\overline{A}+B) =\underbrace{\overline{A}\overline{A}}_{\overline{A}}+\overline{A}B+\overline{A}\overline{B}+\underbrace{\overline{B}B}_{0}$$

$$=\overline{A}+\overline{A}B+\overline{A}\overline{B}$$

$$=\overline{A}(\underbrace{1+B+\overline{B}}_{1})$$

$$=\overline{A} \;//$$

87

$$\boxed{A\overline{B}C+B\overline{C}\overline{D}+ACD+A\overline{B}}$$

$$A\overline{B}C+B\overline{C}\overline{D}+ACD+A\overline{B}$$

$$=A\overline{B}(\underbrace{C+1}_{1})+B\overline{C}\overline{D}+ACD$$

$$=A\overline{B}+B\overline{C}\overline{D}+ACD \;//$$

88

$$\boxed{(A+B+\overline{C})(\overline{A}\overline{B}+C)}$$

$$(A+B+\overline{C})(\overline{A}\overline{B}+C)$$

$$=\underbrace{A\overline{A}\overline{B}}_{0}+AC+\underbrace{B\overline{A}\overline{B}}_{0}+BC+\overline{C}\overline{A}\overline{B}+\underbrace{C\overline{C}}_{0} \quad =AC+BC+\overline{A}\overline{B}\overline{C} \;//$$

(89)

$$\boxed{(\overline{A}+B)\left[\overline{A}\,(B+A)\right]}$$

$(\overline{A}+B)\left[\overline{A}\,(B+A)\right]$

$= (\overline{A}+B)\,(\overline{A}B+\underbrace{\overline{A}A}_{0})$

$= (\overline{A}+B)\cdot\overline{A}B$

$= \overline{A}\cancel{A}B+\overline{A}B\cancel{B}$

$= \overline{A}B+\overline{A}B=\overline{A}B$ //

(90)

$$\boxed{A\overline{B}+\overline{B}\,\overline{C}+\overline{A}\,\overline{C}}$$

$A\overline{B}+\overline{B}\,\overline{C}+\overline{A}\,\overline{C}$

$= A\overline{B}C+A\overline{B}\,\overline{C}+\cancel{A\overline{B}\,\overline{C}}+\overline{A}\,\overline{B}\,\overline{C}+\overline{A}B\overline{C}+\cancel{\overline{A}\,\overline{B}\,\overline{C}}$

(standart çarpımların toplamına dönüştürdük)

$= A\overline{B}C+A\overline{B}\,\overline{C}+\overline{A}\,\overline{B}\,\overline{C}+\overline{A}B\overline{C}$

$= A\overline{B}C+\overline{A}B\overline{C}+\overline{B}\,\overline{C}(\underbrace{\overline{A}+A}_{1}) = A\overline{B}C+\overline{A}B\overline{C}+\overline{B}\,\overline{C}$

$= A\overline{B}C+\overline{C}(\underbrace{\overline{A}B+\overline{B}}_{\overline{B}+A}) = A\overline{B}C+\overline{C}\,\overline{B}+\overline{A}\,\overline{C}$

$= \overline{B}(\underbrace{AC+\overline{C}}_{\overline{C}+A})+\overline{A}\,\overline{C} = \overline{B}\,\overline{C}+\overline{B}A+\overline{A}\,\overline{C}$ aynısı çıktı.

(91)

$$\boxed{\overline{AB+\overline{C}D}}$$

$\overline{AB+\overline{C}D}$

$= \overline{AB}\cdot\overline{\overline{C}D} = (\overline{A}+\overline{B})\cdot(\overline{\overline{C}}+\overline{D})$

$= (\overline{A}+\overline{B})\,(C+\overline{D}) = \overline{A}C+\overline{A}\,\overline{D}+\overline{B}C+\overline{B}\,\overline{D}$ //

(92)

$$\boxed{\overline{\overline{C}\,(\overline{A}+B)}} = \overline{\overline{C}\,(\overline{A}+B)}$$

$= \overline{\overline{C}}+\overline{(\overline{A}+B)}$

$= C+\overline{\overline{A}}\cdot\overline{B} = C+A\overline{B}$ //

(93)

$$\boxed{\overline{(\overline{A}+C)\,(B+\overline{D})}}$$

$\overline{(\overline{A}+C)\,(B+\overline{D})}$

$= \overline{(\overline{A}+C)}+\overline{(B+\overline{D})}$

$= \overline{\overline{A}}\cdot\overline{C}+\overline{B}\cdot\overline{\overline{D}}$

$= A\overline{C}+\overline{B}D$ //

(94)

$$\boxed{\overline{ABC}+\overline{(\overline{D}+E)}}$$

$\overline{ABC}+\overline{(\overline{D}+E)} = (\overline{A}+\overline{B}+\overline{C})\cdot\overline{\overline{D}}\cdot\overline{E}$

$= \overline{A}D\overline{E}+\overline{B}D\overline{E}+\overline{C}D\overline{E}$ /

(95)

$$\boxed{\overline{(\overline{A}+C)\;\overline{AB}}}$$

$\overline{(\overline{A}+C)\;\overline{AB}}$

$= \overline{(\overline{A}+C)}+\overline{\overline{AB}}$

$= \overline{\overline{A}}\cdot\overline{C}+AB$

$= A\overline{C}+AB$ //

(96)

$$\boxed{\overline{\overline{A}B+A\overline{B}}}$$

$\overline{\overline{A}B+A\overline{B}}$

$= \overline{\overline{A}B}\cdot\overline{A\overline{B}}$

$= (\overline{\overline{A}}+\overline{B})\cdot(\overline{A}+\overline{\overline{B}})$

$= (A+\overline{B})\cdot(\overline{A}+B)$

$= \underbrace{A\overline{A}}_{0}+AB+\overline{A}\,\overline{B}+\underbrace{B\overline{B}}_{0}$

$= AB+\overline{A}\,\overline{B}$ //

(97)

$$\boxed{\overline{(AB+\overline{C})\cdot D}}$$

$\overline{(AB+\overline{C})\cdot D}$

$= \overline{ABD+\overline{C}D} = \overline{ABD}\cdot\overline{\overline{C}D}$

$= \overline{ABD}\cdot(\overline{\overline{C}}+\overline{D}) = (\overline{A}+\overline{B}+\overline{D})\cdot(C+\overline{D})$

$= \overline{A}C+\overline{A}\,\overline{D}+\overline{B}C+\overline{B}\,\overline{D}+\overline{D}C+\underbrace{\overline{D}\,\overline{D}}_{\overline{D}}$

$= \overline{A}C+\overline{B}C+\overline{D}(\underbrace{\overline{A}+\overline{B}+C+1}_{1})$

$= \overline{A}C+\overline{B}C+\overline{D}$ //

(98)

$$\boxed{xz+z(\bar{x}+xy)}$$

$$xz+z(\bar{x}+xy)$$

$$=xz+\bar{x}z+xyz$$

$$=xz(\underbrace{1+y}_{1})+\bar{x}z$$

$$=xz+\bar{x}z=z(\underbrace{x+\bar{x}}_{1})$$

$$=z\ //$$

(100)

$$\boxed{\bar{A}+\bar{B}+\bar{A}\bar{B}}$$

$$\bar{A}+\bar{B}+\bar{A}\bar{B}$$

$$=\bar{A}+\bar{B}(\underbrace{1+\bar{A}}_{1})$$

$$=\bar{A}+\bar{B}\ //$$

(103)

$$\boxed{\overline{(A+BC)}+A(B+\bar{C})}$$

$$=\overline{(A+BC)}+AB+A\bar{C}$$

$$=\bar{A}\cdot\overline{BC}+AB+A\bar{C}$$

$$=\bar{A}(\bar{B}+\bar{C})+AB+AC$$

$$=\bar{A}\bar{B}+\bar{A}\bar{C}+AB+AC\ //$$

(105)

$$\boxed{\bar{B}\left[C(\bar{D}+E+C)+\bar{F}C\right]}$$

$$\bar{B}\left[C(\bar{D}+E+C)+\bar{F}C\right]$$

$$=\bar{B}\left[C\bar{D}+CE+\underbrace{CC}_{C}+\bar{F}C\right]$$

$$=\bar{B}\left[C(\underbrace{\bar{D}+E+1+\bar{F}}_{1})\right]$$

$$=\bar{B}C\ //$$

(99)

$$\boxed{DE+\bar{D}+\bar{E}}$$

$$\underbrace{\underset{\bar{x}}{D}\underset{y}{E}+\underset{x}{\bar{D}}}_{\bar{D}+E}+\bar{E}=\bar{D}+\underbrace{E+\bar{E}}_{1}$$

$$=\bar{D}+1=1\ //$$

(101)

$$\boxed{AB+ABC+ABCD+ABCDE+ABCDEF}$$

$$=AB(\underbrace{1+C+CD+CDE+CDEF}_{1})=AB\ //$$

(102)

$$\boxed{A\bar{B}+AB+BC}$$

$$A\bar{B}+AB+BC=A(\underbrace{\bar{B}+B}_{1})+BC=A+BC\ //$$

(104)

$$\boxed{\bar{A}A\bar{C}+\bar{A}B\bar{C}+\bar{A}C\bar{C}+BA\bar{C}+BB\bar{C}+BC\bar{C}}$$

$$\underbrace{\bar{A}A}_{0}\bar{C}+\bar{A}B\bar{C}+\bar{A}\underbrace{C\bar{C}}_{0}+BA\bar{C}+\cancel{B}B\bar{C}+B\underbrace{C\bar{C}}_{0}$$

$$=\bar{A}B\bar{C}+BA\bar{C}+B\bar{C}=B\bar{C}(\underbrace{\bar{A}+A+1}_{1})=B\bar{C}\ //$$

(106)

$$\boxed{\overline{\overline{A+BC}+A\bar{B}}}$$

$$\overline{\overline{A+BC}+A\bar{B}}$$

$$=\overline{\overline{A+BC}}\cdot\overline{A\bar{B}}$$

$$=(A+BC)(\bar{A}+\bar{\bar{B}})$$

$$=(A+BC)(\bar{A}+B)$$

$$=\underbrace{A\bar{A}}_{0}+AB+\bar{A}BC+B\cancel{B}C$$

$$=AB+\bar{A}BC+BC=AB+BC(\underbrace{\bar{A}+1}_{1})$$

$$=AB+AC\ //$$

107

$$\boxed{(A+\bar{B}+C)\,\overline{(AB+\bar{A}\bar{C})}}$$

$$(A+\bar{B}+C)\,\overline{(AB+\bar{A}\bar{C})}$$

$$=(A+\bar{B}+C)(\overline{AB}\cdot\overline{\bar{A}\bar{C}})$$

$$=(A+\bar{B}+C)(\bar{A}+\bar{B})\cdot(\cancel{\bar{\bar{A}}}+\cancel{\bar{\bar{C}}})$$

$$=(A+\bar{B}+C)(\bar{A}+\bar{B})(A+C)$$

$$=(\underbrace{A\bar{A}}_{0}+A\bar{B}+\bar{A}\bar{B}+\bar{B}\cancel{\bar{B}}+\bar{A}C+\bar{B}C)(A+C)$$

$$=(A\bar{B}+\bar{A}\bar{B}+\bar{B}+\bar{A}C+\bar{B}C)(A+C)$$

$$=[\bar{B}(\underbrace{A+\bar{A}+1+\bar{C}}_{1})+\bar{A}C](A+C)$$

$$=(\bar{B}+\bar{A}C)\cdot(A+C)=A\bar{B}+\underbrace{A\bar{A}}_{0}C+\bar{B}C+\bar{A}C\cancel{C}$$

$$=A\bar{B}+\bar{B}C+\bar{A}C\ //$$

108

$$\boxed{\overline{(A+B+C)\cdot D}}$$

$$\overline{(A+B+C)\cdot D}=\overline{(A+B+C)}\cdot\bar{D}$$

$$=\bar{A}\bar{B}\bar{C}\bar{D}\ //$$

109

$$\boxed{\overline{(AB+C)(A+BC)}}$$

$$=\overline{(AB+C)}+\overline{(A+BC)}$$

$$=\overline{AB}\cdot\bar{C}+\bar{A}\cdot\overline{BC}$$

$$=(\bar{A}+\bar{B})\bar{C}+\bar{A}(\bar{B}+\bar{C})$$

$$=\bar{A}\bar{C}+\bar{B}\bar{C}+\bar{A}\bar{B}+\cancel{\bar{A}\bar{C}}$$

$$=\bar{A}\bar{C}+\bar{B}\bar{C}+\bar{A}\bar{B}\ //$$

110

$$\boxed{\overline{(\bar{A}+B)+CD}}$$

$$\overline{(\bar{A}+B)+CD}$$

$$=\overline{\bar{A}+B}\cdot\overline{CD}$$

$$=\bar{\bar{A}}\cdot\bar{B}\cdot(\bar{C}+\bar{D})$$

$$=A\bar{B}(\bar{C}+\bar{D})$$

$$=A\bar{B}\bar{C}+A\bar{B}\bar{D}\ //$$

111

$$\boxed{xy+x(y+z)+y(y+z)}$$

$$=xy+x\cancel{y}+xz+y\cancel{y}+yz$$

$$=xy+xz+y+yz$$

$$=y(\underbrace{x+1+z}_{1})+xz$$

$$=y+xz\ //$$

112

$$\boxed{xy(\bar{x}+y)(\bar{x}+x)}$$

$$xy(\bar{x}+y)(\underbrace{\bar{x}+x}_{1})$$

$$=xy(\bar{x}+y)=xy\bar{x}+xy\cancel{y}$$

$$=\underbrace{x\bar{x}}_{0}y+xy=xy\ //$$

113

$$\boxed{[A\bar{B}(C+\bar{B}D)+\bar{A}\bar{B}]\cdot C}$$

$$=(A\bar{B}C+A\bar{B}\cancel{\bar{B}}D+\bar{A}\bar{B})\cdot C$$

$$=A\bar{B}C\cancel{C}+A\bar{B}CD+\bar{A}\bar{B}C$$

$$=A\bar{B}C+A\bar{B}CD+\bar{A}\bar{B}C$$

$$=A\bar{B}C(\underbrace{1+D}_{1})+\bar{A}\bar{B}C=A\bar{B}C+\bar{A}\bar{B}C$$

$$=\bar{B}C(\underbrace{A+\bar{A}}_{1})=\bar{B}C\ //$$

114

$$\boxed{[A\bar{B}(C+BD)+\bar{A}\bar{B}]\cdot C}$$

$$=(A\bar{B}C+A\underbrace{\bar{B}B}_{0}D+\bar{A}\bar{B})\cdot C$$

$$=(A\bar{B}C+\bar{A}\bar{B})C=A\bar{B}C\cancel{C}+\bar{A}\bar{B}C$$

$$=A\bar{B}C+\bar{A}\bar{B}C=\bar{B}C(\underbrace{A+\bar{A}}_{1})$$

$$=\bar{B}C\ //$$

(115) $\boxed{\overline{(A\overline{B}+\overline{A}B)}\,(A+B)}$

$=\overline{A\overline{B}}\cdot\overline{\overline{A}B}\cdot(A+B)$

$=(\overline{A}+\overline{\overline{B}})\cdot(\overline{\overline{A}}+\overline{B})(A+B)$

$=(\overline{A}+B)(A+\overline{B})(A+B)$

$=(\underbrace{\overline{A}A}_0+\overline{A}\,\overline{B}+AB+\underbrace{B\overline{B}}_0)(A+B)$

$=(\overline{A}\,\overline{B}+AB)(A+B)$

$=\underbrace{\overline{A}A}_0\overline{B}+\overline{A}\underbrace{\overline{B}B}_0+AB\cancel{A}+AB\cancel{B}$

$=AB+AB=AB$ //

(116) $\boxed{\overline{\overline{(A+B)}+\overline{C}}}$

$=\overline{\overline{A+B}}\cdot\overline{\overline{C}}$

$=(A+B)\cdot C$

$=AC+BC$ //

(117) $\boxed{\overline{\overline{(A\overline{B})}\,(\overline{B}+C)}}$

$=\overline{\overline{A\overline{B}}}+\overline{(\overline{B}+C)}$

$=A\overline{B}+\overline{\overline{B}}\cdot\overline{C}$

$=A\overline{B}+B\overline{C}$ //

(118) $\boxed{\overline{(A+B+C)}+\overline{\overline{D}E}}$

$=\overline{A}\,\overline{B}\,\overline{C}+(\overline{\overline{D}}+\overline{E})$

$=\overline{A}\,\overline{B}\,\overline{C}+D+\overline{E}$ //

(119) $\boxed{\overline{(A+B)}+\overline{A}(B+C)}$

$=\overline{A}\,\overline{B}+\overline{A}B+\overline{A}C$

$=\overline{A}(\underbrace{\underbrace{\overline{B}+B}_1+C}_1)=\overline{A}$ //

(120) $\boxed{\overline{(A+B)}\cdot\overline{C}\,(C+D)}$

$=\overline{A}\,\overline{B}\cdot(\underbrace{\overline{C}C}_0+CD)$

$=\overline{A}\,\overline{B}+CD$ //

(121) $\boxed{(\overline{A}+B)C+ABC}$

$=\overline{A}C+BC+ABC$

$=\overline{A}C+BC(\underbrace{1+A}_1)$

$=\overline{A}C+BC$ //

(122) $\boxed{\overline{AB+AC}+\overline{A}\,\overline{B}C}$

$=\overline{AB}\cdot\overline{AC}+\overline{A}\,\overline{B}C$

$=(\overline{A}+\overline{B})(\overline{A}+\overline{C})+\overline{A}\,\overline{B}C$

$=\underbrace{\overline{A}\overline{A}}_0+\overline{A}\,\overline{C}+\overline{A}\,\overline{B}+\overline{B}\,\overline{C}+\overline{A}\,\overline{B}C$

$=\overline{A}\,\overline{C}+\overline{A}\,\overline{B}+\overline{B}\,\overline{C}+\overline{A}\,\overline{B}C$

$=\overline{A}\,\overline{C}+\overline{B}\,\overline{C}+\overline{A}\,\overline{B}(\underbrace{1+C}_1)$

$=\overline{A}\,\overline{C}+\overline{B}\,\overline{C}+\overline{A}\,\overline{B}$ //

(123) $\boxed{(A+\overline{A})(AB+AB\overline{C})}$

$\underbrace{(A+\overline{A})}_1(AB+AB\overline{C})$

$=AB+AB\overline{C}$

$=AB(\underbrace{1+\overline{C}}_1)=AB$ //

(124) $\boxed{A\overline{B}C(BD+CDE)+A\overline{C}}$

$=\underbrace{A\underbrace{\overline{B}B}_0CD}_0+A\overline{B}C\cancel{C}DE+A\overline{C}$

$=A\overline{B}CDE+A\overline{C}=A(\overline{B}CDE+\overline{C})$

$=A(\underbrace{\underbrace{\overline{C}}_{x}+\underbrace{C}_{\overline{x}}\,\underbrace{\overline{B}DE}_{y}}_{x+\overline{x}y=x+y})=A(\overline{C}+\overline{B}DE)=A\overline{C}+A\overline{B}DE$ //

(125) $\boxed{(B+BC)(B+\overline{B}C)(B+D)}$

$(B+BC)(B+\overline{B}C)(B+D)$

$=[B(\underbrace{1+C}_1)]\cdot(B+C)(B+D)$

$=B(B+C)(B+D)=(B\cancel{B}+BC)(B+D)$

$=(B+BC)(B+D)=[B(\underbrace{1+C}_1)]\cdot(B+D)=B(B+D)=BB+BD$

$=B+BD=B(\underbrace{1+D}_1)=B$ //

(126)

$$\boxed{\left[A\bar{B}(C+BD)+\bar{A}\bar{B}\right]\cdot C}$$

$$=\left[A\bar{B}C+A\underbrace{\bar{B}B}_{0}D+\bar{A}\bar{B}\right]C$$

$$=(A\bar{B}C+\bar{A}\bar{B})C=A\bar{B}C\cancel{C}+\bar{A}\bar{B}C$$

$$=A\bar{B}C+\bar{A}\bar{B}C=\bar{B}C(\underbrace{A+\bar{A}}_{1})=\bar{B}C//$$

(127)

$$\boxed{A\bar{B}+A\overline{(B+C)}+B\overline{(B+C)}}$$

$$=A\bar{B}+A\bar{B}\bar{C}+\underbrace{B\bar{B}}_{0}\cdot\bar{C}$$

$$=A\bar{B}+A\bar{B}\bar{C}=A\bar{B}(\underbrace{1+\bar{C}}_{1})$$

$$=A\bar{B}//$$

(128)

$$\boxed{\bar{A}BC+A\bar{B}\bar{C}+\bar{A}\bar{B}C+A\bar{B}C+ABC}$$

$$=\bar{A}C(\underbrace{B+\bar{B}}_{1})+A\bar{B}(\underbrace{\bar{C}+C}_{1})+ABC$$

$$=\bar{A}C+A\bar{B}+ABC=\bar{A}C+A(\underset{x}{\bar{B}}+\underset{\bar{x}}{B}\underset{y}{C})$$

$x+\bar{x}y=x+y$ kuralı

$$=\bar{A}C+A(\bar{B}+C)=\bar{A}C+A\bar{B}+AC$$

$$=C(\underbrace{\bar{A}+A}_{1})+A\bar{B}=C+A\bar{B}//$$

(129)

$$\boxed{A\bar{B}C+\bar{A}BC+\bar{A}\bar{B}C}$$

$$=\bar{B}C(\underbrace{A+\bar{A}}_{1})+\bar{A}BC$$

$$=\bar{B}C+\bar{A}BC=C(\underset{x}{\bar{B}}+\underset{\bar{x}}{B}\underset{y}{\bar{A}})$$

$$=C(\bar{B}+\bar{A})=\bar{B}C+\bar{A}C//$$

(130)

$$\boxed{ABC\left[AB+\bar{C}(BC+AC)\right]}$$

$$=ABC\left[AB+B\underbrace{C\bar{C}}_{0}+A\underbrace{C\bar{C}}_{0}\right]$$

$$=ABC\cdot AB=ABC//$$

(131)

$$\boxed{\overline{AB+AC}+\bar{A}\bar{B}C}$$

$$=\overline{AB}+\overline{AC}+\bar{A}\bar{B}C$$

$$=(\bar{A}+\bar{B})(\bar{A}+\bar{C})+\bar{A}\bar{B}C=\bar{A}\cancel{\bar{A}}+\bar{A}\bar{C}+\bar{A}\bar{B}+\bar{B}\bar{C}+\bar{A}\bar{B}C$$

$$=\bar{A}+\bar{A}\bar{C}+\bar{A}\bar{B}+\bar{B}\bar{C}+\bar{A}\bar{B}C$$

$$=\bar{A}(\underbrace{1+\bar{C}+\bar{B}+\bar{B}C}_{1})+\bar{B}\bar{C}=\bar{A}+\bar{B}\bar{C}//$$

(132)

$$\boxed{ABCD+AB\bar{C}+AB\bar{D}+\bar{A}CD+\bar{B}CD}$$

$$=AB(\underbrace{CD+\bar{C}}_{x+\bar{x}y=x+y})+AB\bar{D}+\bar{A}CD+\bar{B}CD$$

$$=AB(\bar{C}+D)+AB\bar{D}+\bar{A}CD+\bar{B}CD$$

$$=AB\bar{C}+\underbrace{ABD+AB\bar{D}}_{AB}+\bar{A}CD+\bar{B}CD$$

$$=AB(\underbrace{\bar{C}+1}_{1})+\bar{A}CD+\bar{B}CD=AB+\bar{A}CD+\bar{B}CD//$$

(133)

$$\boxed{(AB+C)(\bar{B}D+\bar{C}\bar{E})+\overline{(AB+C)}}$$

$$=A\underbrace{B\bar{B}}_{0}D+AB\bar{C}\bar{E}+\bar{B}CD+\underbrace{C\bar{C}}_{0}\bar{E}+\overline{AB}\cdot\bar{C}$$

$$=AB\bar{C}\bar{E}+\bar{B}CD+(\bar{A}+\bar{B})\cdot\bar{C}$$

$$=AB\bar{C}\bar{E}+\bar{B}CD+\bar{A}\bar{C}+\bar{B}\bar{C}$$

$$=\bar{C}(AB\bar{E}+\bar{A})+\bar{B}(CD+\bar{C})$$

$$=\bar{C}(\bar{A}+B\bar{E})+\bar{B}(\bar{C}+D)$$

$$=\bar{A}\bar{C}+B\bar{C}\bar{E}+\bar{B}\bar{C}+\bar{B}D//$$

(134) $\boxed{[A+\bar{B}C+D+EF]\,[A+\bar{B}C+\overline{(D+EF)}]}$

$=(A+\bar{B}C+D+EF)\left[A+\bar{B}C+\bar{D}\cdot\overline{EF}\right]$

$=(A+\bar{B}C+D+EF)\left[A+\bar{B}C+\bar{D}(\bar{E}+\bar{F})\right]$

$=(A+\bar{B}C+D+EF)(A+\bar{B}C+\bar{D}\bar{E}+\bar{D}\bar{F})$

$=A\cancel{A}+A\bar{B}C+A\bar{D}\bar{E}+A\bar{D}\bar{F}+\cancel{A\bar{B}}C+\bar{B}C\cancel{\bar{B}C}+\bar{B}C\bar{D}\bar{E}+\bar{B}C\bar{D}\bar{F}+AD+\bar{B}CD+\underbrace{D\bar{D}\bar{E}}_{0}+\underbrace{D\bar{D}\bar{F}}_{0}$
$+AEF+\bar{B}CEF+\underbrace{E\bar{E}F\bar{D}}_{0}+\underbrace{EF\bar{D}\bar{F}}_{0}$

$=A+A\bar{B}C+A\bar{D}\bar{E}+A\bar{D}\bar{F}+\bar{B}C+\bar{B}C\bar{D}\bar{E}+\bar{B}C\bar{D}\bar{F}+AD+\bar{B}CD+AEF)+\bar{B}CEF$

$=A(\underbrace{1+\bar{B}C+\bar{D}\bar{E}+\bar{D}\bar{F}+D+EF}_{1})+\bar{B}C(\underbrace{1+\bar{D}\bar{E}+\bar{D}\bar{F}+D+EF}_{1})=A+\bar{B}C$ //

-veya-

$=(\underbrace{A+\bar{B}C}_{x}+\underbrace{D+EF}_{y})\left[\underbrace{A+\bar{B}C}_{x}+\underbrace{\overline{(D+EF)}}_{\bar{y}}\right]=(x+y)(x+\bar{y})=x\cancel{x}+x\bar{y}+yx+\underbrace{y\bar{y}}_{0}$

$=x+x\bar{y}+xy$

$=x(\underbrace{1+\bar{y}+y}_{1})=x$

yani $=A+\bar{B}C$ //

(135) $\boxed{\overline{\overline{A+BC}+\overline{A\bar{B}}}}$

$=\cancel{\overline{\overline{A+BC}}}\cdot\cancel{\overline{\overline{A\bar{B}}}}$

$=(A+BC)(A\bar{B})$

$=A\cancel{A}\bar{B}+A\underbrace{\bar{B}B}_{0}C$

$=A\bar{B}$ //

(136) $\boxed{\overline{(A+B)}\;\overline{(C+D+E)}+\overline{(A+B)}}$

$\underbrace{\overline{(A+B)}}_{x}\;\underbrace{\overline{(C+D+E)}}_{y}+\underbrace{\overline{(A+B)}}_{x}$

$=xy+x=x(\underbrace{y+1}_{1})=x$

$=\overline{A+B}=\bar{A}\bar{B}$ //

(137) $\boxed{\overline{A\bar{B}+\bar{C}D+EF}}=\overline{A\bar{B}}\cdot\overline{\bar{C}D}\cdot\overline{EF}=(\bar{A}+\cancel{\bar{\bar{B}}})(\cancel{\bar{\bar{C}}}+\bar{D})(\bar{E}+\bar{F})$

$=(\bar{A}+B)(C+\bar{D})(\bar{E}+\bar{F})$ //

(138)

$$\boxed{\overline{(A+B)\,\bar C\bar D + E + \bar F}}$$

$$= \overline{(A+B)\,\bar C\bar D} \cdot \bar E \cdot \bar{\bar F}$$

$$= \left[\overline{(A+B)} + \bar{\bar C} + \bar{\bar D}\right] \cdot \bar E F$$

$$= (\bar A\bar B + C + D)\,\bar E F$$

$$= \bar A\bar B\bar E F + C\bar E F + D\bar E F$$

(139)

$$\boxed{\overline{(A+B)} \cdot \bar C}$$

$$= \bar A \cdot \bar B \cdot \bar C$$

(140)

$$\boxed{(\bar A + ABC)(A+\bar C)}$$

$$(\underset{x}{\bar A} + \underset{\bar x}{A}\underset{y}{BC})(A+\bar C)$$

$x+\bar x y = x+y$ kuralı

$$= (\bar A + BC)(A+\bar C)$$

$$= \underbrace{\bar A A}_{0} + \bar A\bar C + ABC + \underbrace{BC\bar C}_{0}$$

$$= \bar A\bar C + ABC$$

(141)

$$\boxed{\overline{\bar A\bar B\bar C} + \overline{\bar A\bar B\bar C \cdot C} + D}$$

$$= (\bar{\bar A} + \bar{\bar B} + \bar{\bar C}) + (\bar{\bar A} + \bar{\bar B} + \bar{\bar C} + \bar C) + D$$

$$= (A+B+C) + (A+B+\underbrace{C+\bar C}_{1}) + D$$

$$= A+B+C+A+B+1+D = 1$$

(142)

$$\boxed{AB + A(B+C) + B(B+C)}$$

$$= AB + \cancel{AB} + AC + B\cancel{B} + BC$$

$$= AB + AC + B + BC$$

$$= B(\underbrace{A+1+C}_{1}) + AC = B + AC$$

(143)

$$\boxed{\overline{\overline{\overline{A+B}+C} + \overline{\overline{D+E}+F}}}$$

$$= \overline{\overline{\overline{A+B}+C}} \cdot \overline{\overline{\overline{D+E}+F}}$$

$$= (\overline{A+B}+C) \cdot (\overline{D+E}+F)$$

$$= (\bar A\bar B + C) \cdot (\bar D\bar E + F)$$

$$= \bar A\bar B\bar D\bar E + \bar A\bar B F + C\bar D\bar E + CF$$

(144)

$$\boxed{\bar A\bar B\bar C + A(C\bar D + \bar B)}$$

$$= \bar A\bar B\bar C + AC\bar D + A\bar B$$

$$= \bar B(\underbrace{\underset{\bar x}{\bar A}\underset{y}{\bar C} + \underset{x}{A}}) + AC\bar D$$

$x+\bar x y = x+y$ kuralı

$$= \bar B(A+\bar C) + AC\bar D$$

$$= A\bar B + \bar B\bar C + AC\bar D$$